Fiscal Year:	FY 2021	**Task Last Updated:**	FY 08/30/2021
PI Name:	Blancaflor, Elison B. Ph.D.		
Project Title:	Leveraging Spaceflight Genomic Data to Uncover Developmental and Cell Type Specific Gene Regulatory Networks in Plants Responding to Gravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential (2) Bioregenerative Life Support		
PI Email:	elison.b.blancaflor@nasa.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	321-501-5368
Organization Name:	NASA Kennedy Space Center		
PI Address 1:	Utilization and Life Science Office		
PI Address 2:	Exploration and Research Technology Programs		
PI Web Page:			
City:	Kennedy Space Center	**State:**	FL
Zip Code:	32955	**Congressional District:**	8
Comments:	NOTE: Dr. Blancaflor moved to NASA Kennedy Space Center in spring 2021; as a PI, his prior grants were for work at the Noble Foundation in Oklahoma.		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	02/01/2019	**End Date:**	07/31/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:	Ruby, Anna Maria	**Contact Phone:**	321-867-7065
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2021 per transfer of project to Dr. Simon Gilroy, University of Wisconsin (Ed., 9/21/21)		
Key Personnel Changes/Previous PI:	Final report September 2021: Dr. Sabrina Chin is a postdoctoral fellow who is supported on this grant. She started on September 9, 2019. However, the COVID-19 pandemic continued to hinder progress on this grant due to limited access to laboratory facilities. In addition to COVID-19, the Noble Research Institute leadership discontinued its programs on plant science, plant breeding and laboratory-based science research. A fallout from this decision was the termination of all external grants that were not aligned with the new focus of the Institute on regenerative ranching. This NASA grant was one of many federal and state grants at the Noble Research Institute that were asked to be discontinued. Furthermore, the PI (Blancaflor) and Co-Is (Tang and Kwon) ended their employment at the Noble Research Institute as a result of the decision to terminate basic plant science research. Therefore, on March 2021, a request was made to NASA to transfer remaining work on this grant to Dr. Simon Gilroy at the University of Wisconsin. PI change: Dr. Simon Gilroy will be the new PI at Wisconsin) Other key personnel change Dr. Sabrina Chin (former postdoctoral fellow at Noble who will continue as a postdoc in the Gilroy lab at Wisconsin).		

COI Name (Institution):	Kwon, Taegun Ph.D. (Noble Foundation) Tang, Yuhong Ph.D. (Noble Foundation)
Grant/Contract No.:	80NSSC19K0129
Performance Goal No.:	
Performance Goal Text:	
Task Description:	As NASA pursues its goal of sending human colonies to Mars and beyond, there is a pressing need to better understand how the spaceflight environment influences fundamental biological processes. Understanding how plants respond to gravity on Earth and the microgravity environment in space is of particular interest because of the important role that plants will play in advanced bioregenerative life support systems during long duration space missions. In recent years, numerous transcriptomic datasets have shown that hundreds of genes change expression when plants are grown in space. However, it has been challenging to tease apart which genes change due to microgravity from those that change as a result of general plant stress responses. This difficulty stems from the fact that plant spaceflight genomics datasets have been derived almost exclusively from whole seedlings and plant organs, which could potentially mask or dilute gene expression changes that are occurring within the distinct cell-types that are responsible for gravity sensing (e.g., root columella and stem endodermal cells). Thus, the major goal of this NASA-ground-based proposal is to begin to take steps to develop methodology that will enable space genomic studies on homogenous cell populations. Here, we propose to use laser capture microdissection (LCM) to conduct RNA-Sequencing (RNA-Seq) on gravity sensing columella cells to identify gene regulatory networks that govern early responses of plants to gravity. We hypothesize that gene profile signatures from a homogenous population of columella cells will more clearly reflect transcriptomic changes that are directly linked to microgravity. The RNA-Seq data that we obtain from our analysis of root columella cells will be compared to columella-specific gene modules we identified through our Weighted Gene Co-expression Network Analysis (WGCNA) of Arabidopsis seedlings grown for 6 and 11 days on the International Space Station (ISS) (specific aim 2). In parallel, we will refine LCM methods with the goal of implementing them for future plant spaceflight genomics (specific aim 1). The latter aim will involve ground-based experiments in which specific steps in the LCM methods will be modified to mirror in-orbit operations. Upon successful completion of this proposal, we expect to generate new tools for conducting genomics in homogenous plant cell populations that will enable us to better differentiate microgravity effects from microgravity independent effects. NOTE September 2021: The Principal Investigator (PI) (Blancaflor) and Co-Is (Tang and Kwon) ended their employment at the Noble Research Institute as a result of the Noble Research Institute's decision to terminate basic plant science research. Therefore, on March 2021, a request was made to NASA to transfer remaining work on this grant to Dr. Simon Gilroy at the University of Wisconsin and Dr. Gilroy will be new PI for this work.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research seeks to fill technological gaps that will enable a better understanding of how plants develop in the microgravity environment of spaceflight. The focus of this research is to develop methods that will allow us to learn more about how expression of genes within specific root cell types change under microgravity or stress-induced spaceflight. The plant organ of interest in this proposal is the root. Because roots are crucial for uptake of water and nutrients to support optimal plant growth, results from this proposal will benefit agriculture on Earth. We expect that genes, which show cell-type expression changes in response to gravity could be potential targets for pre-breeding efforts toward developing crop cultivars that are resilient to environmental stress imposed by marginal lands on Earth and confined growth environments during spaceflight.
	FINAL REPORTING AUGUST 2021 Although progress was made on with this proposal, it was limited because of the continuing COVID-19 pandemic and closure of the PI's laboratory at the Noble Research Institute. Below we summarize progress thus far. A. Root Cap Harvest of Maize Seedlings as Alternative Experiments to Laser Capture Microdissection (LCM) The maize root cap functions as the site of gravity sensing with columella cells that contain amyloplasts that sediment according to the gravity vector. The maize root has a closed type with a clear demarcation boundary between the root cap and the root proper. As such, the root cap can be easily removed. We were able to manually excise the root caps with a scalpel without the aid of a microscope as the root cap was less opaque. We completed the harvest of five biological replicates of Zea mays B73 root caps for our experiment in late January 2021. The RNA integrity number (RIN) based on Bioanalyzer 2100 (Agilent) readings of harvested root cap total RNA were highly satisfactory, ranging from 7-9.6, with concentrations averaging at 15 ng/ul. By using this technique in favor of laser capture microdissection (LCM), we were able to expand our time series to five time points (5 minutes, 10 minutes, 30 minutes, 1 hour and 2 hours after reorienting the root) that relate to important biological responses that included gravity perception, gravity signal transduction and root growth response. In contrast, the LCM technique would have limited us to three time points as this sampling technique is time consuming and laborious. Furthermore, access to the LCM at Noble was limited due to precautions with COVID-19. Whilst enriching for columella cells by harvesting the root cap is less precise than LCM, we expect that the high population of columella cells in the root cap will generate results that are specific enough to columella cell signatures. Three biological replicates of 100 ng root cap total RNA with satisfactory RIN and concentrations were selected for the cDNA library construction. cDNA library was sequenced at 150 base paired reads on Illumina Novaseq S4 at the Oklahoma Medical Research Foundation, Oklahoma City, OK. B. Differential Gene Expression in Maize Root Caps at Various Time Points After Gravistimulation RNA-seq raw reads were aligned to B73 genome (AGPv2) using HiSat2 alignment. The read counts for each gene were based on Zea mays B73 NAM genome version 5 using HTSeq-count tool that processes the counts for differential expression analysis. Differentially expressed genes (DEGs) were analysed based on differences between control and gravistimulated roots using DESeq2 and Limma-voom on the integrated Differential Expression and Pathway analysis (iDEP) platform. Currently, there is no consensus on the softwares used for RNA-Seq analysis. As such, we explored both DESeq2 and Limma-voom to compare the overlapping DEGs. We have also tested the pseudoaligment tool,

Kallisto, and its accompanying DEG analysis tool, Sleuth, but as this technique did not yield overlapping DEGs with DESeq2 and Limma-voom, this is not discussed here.

We interrogated our data structure based on gene counts output from HT-Seq-count using principal component analysis (PCA). We found that our data can be grouped based on the treatment factor, i.e., control or gravistimulated. Notably, 19% (PC2) of our data is statistically significantly correlated to the treatment factor. DESeq2 analysis was set to 2-fold change threshold with a 0.1 false discovery rate cutoff revealed 84 DEGs at 30 minutes and 45 DEGs at 2 hours. On the other hand, Limma-voom analysis using the same 2-fold change threshold and 0.1 FDR cutoff displayed 103 DEGs. To narrow down our list of gene candidates, we focused on DEGs that overlapped between DESeq2 and Limma-voom and reduced the number of DEGs obtained from DESeq2 by restricting the fold change threshold to 4. Next, we cross-referenced the list of 74 DEGs from DESeq2 with the 103 DEGs from Limma-voom analysis and found 19 corroborating DEGs. In this group of 19 genes, nine were mostly upregulated throughout the five time points, and the remaining ten were mostly downregulated. In the group of genes that were upregulated, there was an interesting trend whereby genes that were initially upregulated at 5 minutes showed reduced expression at 10 minutes. Gene expression trends from DESeq2 analysis will be further verified by a selection of genes for quantitative (q) reverse transcriptase (RT)-polymerase chain reaction (PCR). We have selected seven genes from this list of 19 overlapping genes and included additional seven genes based on their postulated function that showed differential expression in DESeq2 analysis. Genes that were of interest have plausible functions pertaining to gravity reorientation growth, such as ion channels for the transduction of gravity signal, metabolism relating to glucogenesis and respiration to power growth, cell wall and cytoskeleton related to orchestrate changes in cell shape for growth and hormone biosynthesis that includes auxin, cytokinin, and gibberellin.

NOTE: The PI (Blancaflor) and Co-Is (Tang and Kwon) ended their employment at the Noble Research Institute as a result of the Noble Research Institute's decision to terminate basic plant science research. Therefore, on March 2021, a request was made to NASA to transfer remaining work on this grant to Dr. Simon Gilroy at the University of Wisconsin and Dr. Gilroy will be new PI for this work. See that project, “Leveraging Spaceflight Genomic Data to Uncover Developmental and Cell Type Specific Gene Regulatory Networks in Plants Responding to Gravity (PI: Gilroy)” for future reporting.

Task Progress:

ANNUAL REPORTING NOVEMBER 2020

Impact of COVID-19 on Task Progress

Our progress for this reporting period was adversely affected by the COVID-19 pandemic. On March 19, 2020, 90% of Noble Research Institute employees were required to work from home to minimize the spread of the virus. The 10% allowed to work on-campus were limited to essential personnel and did not include the Principal Investigator (PI), Co-Is, and postdoctoral fellow. On June 15, 2020, 50% of employees were allowed to return to campus and such restrictions continue to date because of the on-going threat of the disease. During this phase in which 50% of employees are allowed on-campus, personnel involved in this project have limited access to laboratory facilities. This allowed progress in accomplishing specific aims of the project, but as noted below, achieving such progress required us to consider alternative experiments.

Gene Expression Changes in Specific Gravity Sensing Cells of Roots

Prior to campus closure in March, we managed to evaluate RNA integrity numbers (RINs) from maize columella cells harvested using laser capture microdissection (LCM). RIN is an indicator of the quality of RNA for planned gene expression studies. Unfortunately, the RNA yield and quality from these samples were poor and we were unable to optimize due to COVID-19-related campus closure. The LCM protocols proposed in the original proposal are technically demanding, and require long hours of trouble shooting in the laboratory. Because of campus closure and limited access to the LCM instrument, we decided to pursue a different approach in obtaining cell-type specific transcripts that are differentially regulated by gravity. This approach involved dissecting root caps from maize roots. The experimental approach involved mounting 3-day-old maize (B73 cultivar) seedlings on large square Petri plates lined with moist germination paper so their roots were vertical. After allowing roots to grow vertically for one hour, half of the plates were kept vertical while the other half were rotated by 90 degrees. 5 min, 10 min, 30 min, 1 h, and 2 h after plates were rotated, the tips of 30-35 roots were cut and immediately immersed in RNA Later to preserve RNA. Each gravistimulated time point had its own vertical control. After two hours of incubation in RNA Later, root caps were individually dissected and transferred to a new set of microfuge tubes containing RNA lysis buffer to break open the cells in the harvested root caps. RNA was extracted using a commercially available kit. We found that RNA yield and quality from the harvested root caps was sufficient to begin constructing a library for RNA-Sequencing (RNA-Seq) needed for gene expression studies.

One of advantages of harvesting root caps instead of LCM-derived columella sections is that we were able to expand our sampling time points to five time points that corresponded to gravity perception, gravity signal transduction, and growth response stages. This will enable us to perform hierarchical clustering of genes that are differentially expressed at these times. In addition, we anticipate that this technique is more applicable for future spaceflight experiments, whereby astronauts can harvest root segments containing the root cap into RNA later for further root cap processing on the ground.

Although our efforts during the past 4-5 months have shifted from LCM-isolated maize columella to entire root caps, we still plan to revisit the LCM approach once COVID-19 crisis allows better access to facilities. One specific strategy that we envision to make up for lost time is to validate differentially regulated transcripts from RNA-Seq of harvested maize root caps described above in LCM-dissected columella cells using real-time quantitative-real time RT-PCR. We expect that several transcripts that are differentially regulated by gravistimulation would be specifically expressed in the columella. Cell-Type Specific Expression of Cytosolic Calcium Reporters in Arabidopsis thaliana Roots

For this reporting period, we addressed other research questions that fall within the general theme of cell-type specific responses to gravity. To this end, we evaluated lines of the model plant, Arabidopsis thaliana, that were previously engineered to express a fluorescent cytoplasmic calcium reporter called GCaMP3. Expression of CGaMP3 and other genetically-encoded cytoplasmic calcium sensors in plants are driven by promoters that are expressed throughout the entire plant. These plant lines expressing cytoplasmic calcium sensors in all cell types enable studies of cytoplasmic calcium changes across entire plant organs. However, it is likely that each cell type within the organ is characterized by a unique calcium response. Having lines that express cytoplasmic calcium reporters in specific cell-types would be of particular value to studies of plant gravity responses. Notably, lines that express such reporters in the columella and

	endodermis, which are widely accepted to be the major gravity sensing cells in roots and shoots, respectively, would enable studies that can identify physiological responses that are due primarily to gravity. These columella- and endodermis-specific GCaMP3 lines were published recently. They could potentially be used to evaluate calcium signaling events in microgravity using either the light microscopy module or the spectrum imager on the International Space Station.
Bibliography Type:	Description: (Last Updated: 01/13/2026)
Articles in Peer-reviewed Journals	Krogman W, Sparks JA, Blancaflor EB. "Cell type-specific imaging of calcium signaling in Arabidopsis thaliana seedling roots using GCaMP3." Int J Mol Sci. 2020 Sep 2;21(17):E6385. https://doi.org/10.3390/ijms21176385 ; PMID: 32887481; PMCID: PMC7503278 , Sep-2020
Articles in Peer-reviewed Journals	Chin S, Kwon T, Khan BR, Sparks JA, Mallery EL, Szymanski DB, Blancaflor EB. "Spatial and temporal localization of SPIRRIG and WAVE/SCAR reveal roles for these proteins in actin-mediated root hair development." Plant Cell. 2021 Jul;33(7):2131-48. https://doi.org/10.1093/plcell/koab115 ; PMID: 33881536; PMCID: PMC8364238 , Jul-2021
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Articles in Peer-reviewed Journals	Sparks JA, Sun L, Chin S, Ramanjulu N, Wen J, Gilroy S, Blancaflor EB, Khan BR. "Mutations adjacent to the nucleotide-binding cleft of Arabidopsis thaliana ACTIN7 confer resistance to the actin-disrupting compound latrunculin B." J Exp Bot. 2026 Jan;77(1):212-25. https://doi.org/10.1093/jxb/eraf346 ; PubMed PMID: 40747687 , Jan-2026